Artificial Intelligence

Assignment 8

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1 Environments (8 Points)

Consider these properties:

- completely observable vs. partially observable
- deterministic vs. non-deterministic
- episodic vs. sequential
- static vs. dynamic
- discrete vs. continuous
- single agent vs. multi agents

Name for the following examples which properties for the environments they satisfy and explain briefly why.

a) Minesweeper AI

b) World of Warcraft bot

c) Lawn mower controller

d) Detector for memes in images

a. Minesweeper AI

Partially observable : In the Minesweeper game, the full state of the system is not observable at any point in time. Initially all the blocks will be closed where we don't know which block has been mine. Once we click on the box, we will get the numbers representing information about the mines.

Deterministic : The place of mines in the board are initialized before the start of the game. All the boxes are fixed with either a mine or number, and they are not going to change during the game. Hence it produces the same results every time it is opened.

Sequential : The minesweeper environment is sequential. The agent's current action will affect the future action. When the current block is clicked & if it showing the grid of squares with numbers. These numbers indicate the number of mines touching the current block. Hence the present action is going to affect the future action.

Static : The environment in the minesweeper is not changing. Hence it is static.

Discrete : In the minesweeper game environment the number of actions & percepts are finite to obtain the output. Hence it is Discrete.

Single Agent : In the minesweeper game environment there is only one user clicking the boxes in the grid. Hence it is a single agent.

b. World of Warcraft Bot

Partially observable : It is partially observable as the character has a limited vision and the full state of the system is not observable at any point in time.

Non deterministic: Events occur at random and are dependent on the previous events or choices made by the player or their interactions with elements of the game or other players.

Sequential: Sequential because the character can move in different directions and perform different actions. The surroundings of the character changes based on the previous set of choices made by the player.

Dynamic: The environment changes based on the series of actions taken and previous set of choices made by the player.

Continuous: The number of actions to be performed to obtain the output cannot be determined. The choices/series of actions taken by the character are not limited. Hence it is continuous.

Multi agents : In the world of Warcraft ,there are 4 characters where the bot can take any one of them. Hence it is a multi agent environment.

c. Lawn mower controller

Completely Observable : The environment for the lawn mower is basically the entire space where it has to mow the grass or plant. At any given point in time the area to mow is completely observable.

Sequential : The next of the agent can be dependent on the current incident.

Deterministic : Events donot occur at random. The series of actions taken by the lawn mower and their outputs are predefined and can be determined.

Episodic : The current action is not going to affect a future action. The future actions are not dependent on the previous actions. The Lawn mower controller will mows the grass if it detects.

Static : The environment is not going to change when the agent is deciding on an action.

Continuous : The number of actions to be performed to obtain the output cannot be determined. Hence it is continuous.

Single agent : There are no other agents involved in the environment. The lawn mower will mow the grass/plants

d. Detector for memes in Image

Fully observable : Detector for memes in an image is fully observable because the AI has access to view and analyze the complete image.

Deterministic: Events don't occur at random. The series of actions taken by the meme detector and the output can be determined.

Episodic: The detector performs analysis without the influence of outer environment, and each analysis can be considered an episode. Hence episodic.

Static: The image does not change when the detector is performing analysis.

Discrete: The number of steps taken to detect a meme and to obtain the output are finite. Hence it is Discrete.

Single agent: There are no other agents involved in the environment other than the detector.

2 Reflex-based and BDI agents (12 Points)

A parcel delivery drone is supposed to transport parcels from the postal service's depot (D) to customers ($C_1, ..., C_{100}$). It can pick up a parcel (but not more than one) and drop it again. It can also fly to any customer or to the depot. Whenever it is at the depot, its batteries are automatically recharged to allow for up to four flights between any two locations. The drone does not have any sensors to measure its remaining charge.

We will represent this drone as an agent. Perceptions for the agent are:

- $location \in \{D, C_1, ..., C_{100}\}$ to indicate the current location of the drone,
- $carries \in \{true, false\}$, and
- $addressee \in \{none, C_1, ..., C_{100}\}$, where $none$ occurs exactly when no parcel it held.

1. Represent the drone through a simple, reflex-based agent. To do so, define the sets P and A and the function act for this agent.
2. Represent the drone through a BDI agent. To do so, define the sets of *Perceptions, Actions, Beliefs, Desires, Intentions.* Make sure that you have fewer intentions than actions, and fewer desires than intentions. Remember that the drone does not have unlimited fuel.

   For one pair of belief and intent, state the plan. Make sure that your plan contains more than one element.

1. A simple reflex based agent.

   The agent is implemented by a single function ,
   act : P -> A
   Where P : set of all possible perceptions
   A : set of all possible actions

   act (l, c, a) =
   {

   pick if l = D and c = false and a ∈ { C1, C2, …., C100 },

   flytoAddress if l = D and c = true and a ∈ { C1, C2, …., C100 },

   flytoDepot if l ∈ { C1, C2, …., C100 } and c = false and a = none,

   drop if l ∈ { C1, C2, …., C100 } and c = True and a∈ { C1, C2, …., C100 } and i = a, chargeBattery if l = D and c= false and a = none ,

}

The strategy is,

- If the current location of the drone is at Depot, which has a delivery addressee in any one of { C1, C2, …., C100 } and does not carry any package, then pick the package.
- If the current location of the delivery drone is at Depot, carries a parcel which has to be delivered to the addressee is any one of { C1, C2, …., C100 } then fly the drone to that address.
- If the current location of the delivery drone is any one of { C1, C2, …., C100 }, carries a parcel which has to be delivered to the addressee is any one of { C1, C2, …., C100 } which is the same as the current location then drops the package.

- If the current location of the drone is any one of the { C1, C2, …., C100 } and it is not carrying any parcel and the addressee held is none, then fly back to Depot
- If the current location of the drone is at Depot,and it is not carrying any parcel and the addressee held is none, then charge the drone.

A = { pick, flytoAddress, flytoDepot, drop, chargeBattery }

P = {

(D,False,a ∈ { C1, C2, …., C100 }),

((D,true,a ∈ { C1, C2, …., C100 }),

(l ∈ { C1, C2, …., C100 },false, held),

(l ∈ { C1, C2, …., C100 },True ,a ∈ { C1, C2, …., C100 } and l = a),

(D, false, held)

}

2. BDI agent

   Perceptions are of form (current location , carries parcel, addressee, battery level)
   Here the drone does not have a sensor to measure the battery level. When the drone is at the depot it will charge automatically up to 4 flights between any two locations. In the battery level we will specify how many levels are left.

   Beliefs are of form (currentlocation, carries parcel, addressee, battery level , flying)

   Desires { deliverPackage, chargeBattery }

   Intentions { deliverPackage, chargeBattery, pickingParcel, dropingParcel }

   Actions are { Pick, Drop, flvtoaddress , flytoDepot, chargeBattery , Idle }

The initial state of agent be,
Bel = (D, “unknown”, “unknown”, 4 , “unknown”)
Des = ø
Int = ø

Let the agent perceives (D, Yes, C1, 2)

brf : P X B -> B
The agent updates the belief using belief recursive brf.
B = (D, Yes, C1, 2, “unknown”)

The agent computes the current options :

The belief of the agent that drone is carrying the package and has a address generates the desire deliverPackage.
Furthermore the battery level is not full, chargeBattery is a desirable goal as well.
Des = { deliverPackage, chargeBattery }

The filter is used to weight against the two options.
As the battery level is 2 which is quite alright, priority is given to the goal deliverPackage.
Int = { deliverPackage}

Now a plan is generated for the selected intention deliverPackage ,

π = (flvtoaddress, Drop)

The plan consists of two actions.

The plan now will be executed.